FYUGP B.Sc. MATHEMATICS HONOURS

ABSTRACT ALGEBRA I (MAT5CJ302)

Semester V · Core in Major

MODULE 2: SUBGROUPS, CYCLIC GROUPS, HOMOMORPHISMS AND PERMUTATIONS

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

1 Define an improper subgroup of a group G. **(1 mark)**

2 State whether true or false: if G has more than one element, then $\{e\}$ is a proper subgroup of G. **(1 mark)**

3 State whether true or false: for a in a group G, the cyclic subgroup $\langle a \rangle$ is the smallest subgroup of G that contains a. **(1 mark)**

4 Find the quotient and remainder when -7 is divided by 3 according to the division algorithm. **(1 mark)**

Section B: 2 marks each, answer all

5 State the division algorithm for $\mathbb{Z}$. **(2 marks)**

6 Write the cycle $(1, 2, 3, 4)$ as a product of transpositions. **(2 marks)**

7 Find all orbits of the permutation $\sigma = (1, 2, 6)(3, 5)$ in S_6. **(2 marks)**

8 Define the right regular representation of a group G. **(2 marks)**

9 Let $H = \{42n + 72m \mid n, m \in \mathbb{Z}\}$. Find the positive generator of H. **(2 marks)**

Section C: 4 marks each, answer any two

10 Let H be a finite nonempty subset of a group G that is closed under the operation of G. Show that H is a subgroup of G. **(4 marks)**

11 Let G be a group and $a \in G$. Show that $\langle a \rangle$ is the smallest subgroup of G containing a. **(4 marks)**

12 Let r and s be positive integers. Show that $H = \{nr + ms \mid n, m \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$. **(4 marks)**

13 Let $\phi : G \to G'$ be a group homomorphism and $a \in G$. Show that $\phi(a^{-1}) = \phi(a)^{-1}$. **(4 marks)**

14 Let G be a group and $g \in G$. Define $\lambda_g : G \to G$ by $\lambda_g(x) = gx$. Show that λ_g is one-to-one and onto G. **(4 marks)**

Section D: 10 marks each, answer any one

15 Let G be a cyclic group with n elements generated by a, and let s be a positive integer. Show that $\langle a^s \rangle$ is a subgroup of G with $\dfrac{n}{\gcd(n, s)}$ elements. **(10 marks)**

16 State and prove Cayley's theorem using the right regular representation rather than the left regular representation. **(10 marks)**

MODULE 2: MODEL ANSWERS

Key steps for checking your work

Section A.

1. An improper subgroup of G is G itself. All other subgroups are proper (Definition 5.2).

2. True. $\{e\}$ is a subgroup (Definition 5.1), and it differs from G when G has more than one element.

3. True. $\langle a\rangle = \{a^n \mid n \in \mathbb{Z}\}$ is a subgroup containing a (Theorem 5.19). Every subgroup containing a contains all a^n, so it contains $\langle a\rangle$.

4. $-7 = 3(-3) + 2$. Quotient -3, remainder 2 (Theorem 6.2).

Section B.

5. For a positive integer m and any integer n there are unique integers q and r with $n = mq + r$ and $0 \leq r < m$ (Theorem 6.2).

6. $(1,2,3,4) = (1,4)(1,3)(1,2)$, read right to left (the formula before Theorem 8.15).

7. The orbits are $\{1,2,6\}$, $\{3,5\}$ and $\{4\}$ (Section 8, orbits before Example 8.18).

8. $\tau : G \to S_G$ with $\tau(x) = \sigma_{x^{-1}}$, where $\sigma_y(g) = gy$ (Definition 8.13). So $\tau(x)(g) = gx^{-1}$.

9. 6. Since $\gcd(42, 72) = 6$ and $6 = 3(72) - 5(42)$, the positive generator of H is 6 (Definition 6.8, Example 6.9).

Section C.

10. For $a \in H$ every power a^k with $k \geq 1$ lies in H. Since H is finite, $a^i = a^j$ for some $i < j$, so $a^{j-i} = e \in H$. Then $a^{-1} = a^{j-i-1} \in H$. So H meets the three conditions of Theorem 5.12, and H is a subgroup (Theorem 5.16).

11. $\langle a\rangle$ is a subgroup containing a (Theorem 5.19). If K is any subgroup containing a, then K contains a^n for every $n \in \mathbb{Z}$, so $\langle a\rangle \subseteq K$.

12. $0 = 0r + 0s \in H$. $(nr + ms) + (n'r + m's) = (n + n')r + (m + m')s \in H$. $-(nr + ms) = (-n)r + (-m)s \in H$. So H is a subgroup (Theorem 5.12). By Corollary 6.7 it is $d\mathbb{Z}$ with $d = \gcd(r, s)$ (Definition 6.8).

13. $\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e) = e'$ (Theorem 8.5, part 1). Multiplying on the left by $\phi(a)^{-1}$ gives $\phi(a^{-1}) = \phi(a)^{-1}$ (Theorem 8.5, part 2).

14. If $\lambda_g(x) = \lambda_g(y)$ then $gx = gy$, so $x = y$ by left cancellation: λ_g is one-to-one. For $b \in G$, $\lambda_g(g^{-1}b) = b$, so λ_g is onto G (Definition 8.10).

15. Let $d = \gcd(n, s)$. The order of a^s is the least $m > 0$ with $n \mid sm$, that is $\frac{n}{d} \mid \frac{s}{d}m$. Since $\gcd(\frac{n}{d}, \frac{s}{d}) = 1$, this holds exactly when $\frac{n}{d} \mid m$. So the order is $\frac{n}{d}$, and $\langle a^s\rangle$ has $\frac{n}{\gcd(n,s)}$ elements (Theorems 5.19, 6.10 and 6.15).

16. For $y \in G$ let $\sigma_y(g) = gy$. Each σ_y is a bijection with inverse $\sigma_{y^{-1}}$. Let $\tau(x) = \sigma_{x^{-1}}$. Then $\tau(xy)(g) = g(xy)^{-1} = gy^{-1}x^{-1} = \tau(x)(\tau(y)(g))$, so τ is a homomorphism into S_G. If $\tau(x) = \tau(y)$, then $x^{-1} = y^{-1}$ (take $g = e$), so $x = y$: τ is one-to-one. Hence $G \cong \tau[G] \leq S_G$ (Definition 8.13, Theorem 8.5, Exercise 54).

MODULE 2: SYLLABUS MAPPING

Topics the past papers do not reach, and the model questions that cover them

Topic	**Book section (8th ed.)**	**Model questions**
Improper and trivial subgroups	5.2	1, 2
Closure test for finite subsets	5.16	10
The subgroup generated by an element, and its minimality	5.19 to 5.20	3, 11
Division algorithm	6.2	4, 5
Greatest common divisor and its subgroup	6.7 to 6.9	9, 12
Order of a power of a generator in a finite cyclic group	6.10 to 6.17	15
Homomorphism properties	8.5	13
Left regular representation	8.10 to 8.11	14
Transpositions	8.14 to 8.15	6
Orbits	8.17 to 8.18	7
Right regular representation and Cayley's theorem	8.13 (Exercise 54)	8, 16